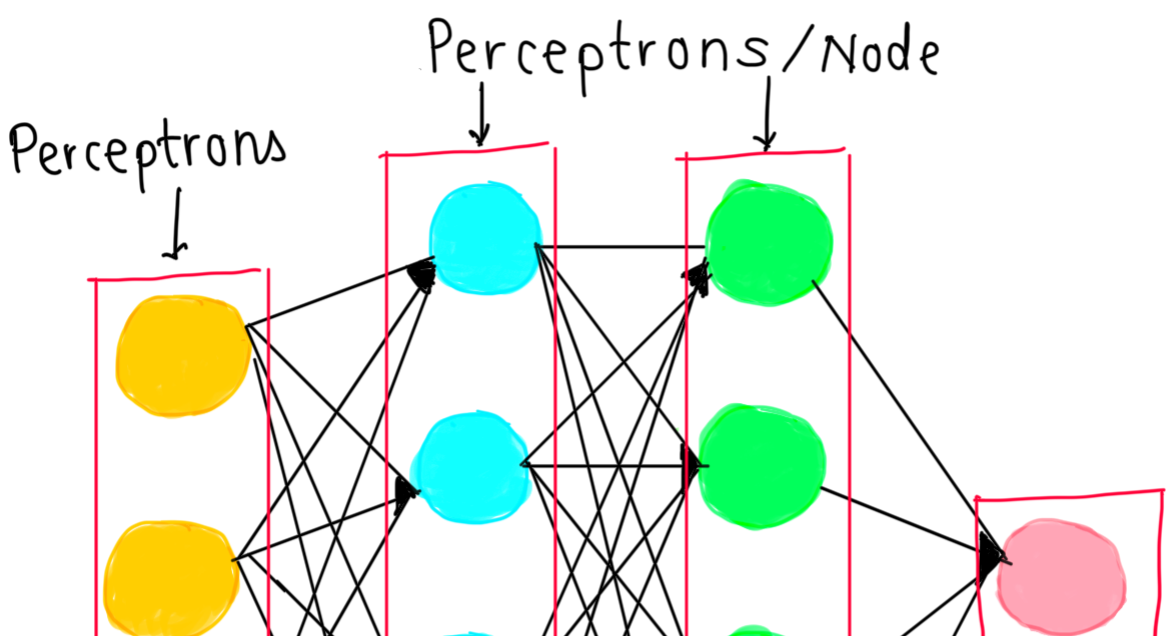
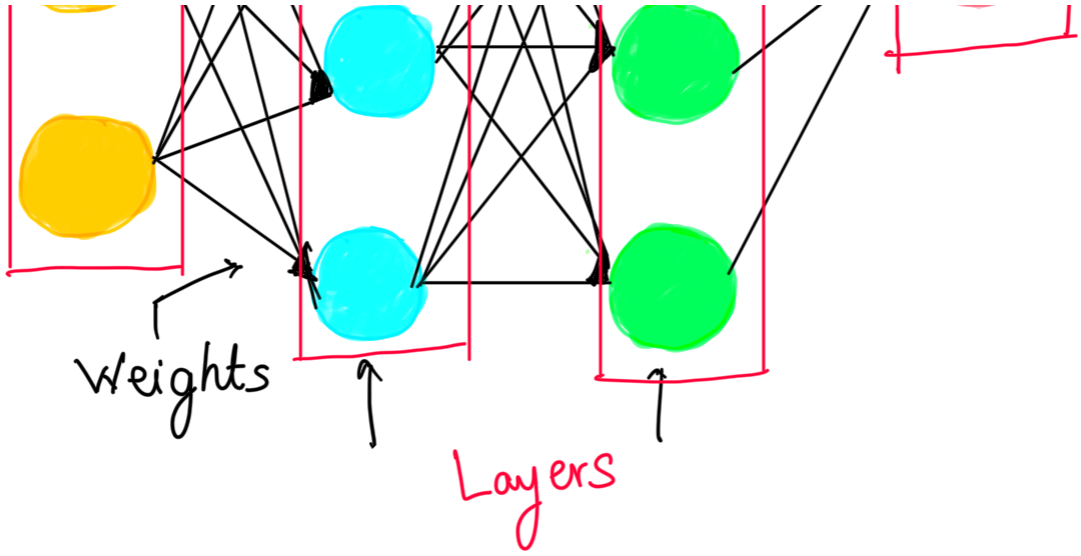
Deep Learning

What is deep learning?

DL is subfield of AI & ML that is inspired by the structure of human brain.

DL algorithms attempt to draw similar conclusions as human would by continually analysing data with a given logical structure called neural network.





Input layer	hidden Layer 1	hidden Layer 2	output Layer

ANN Structure

Deep Learning Definition 1

DL is part of a broader Family of ML methods based on ANN (Artificial Neural Network) with representation learning.

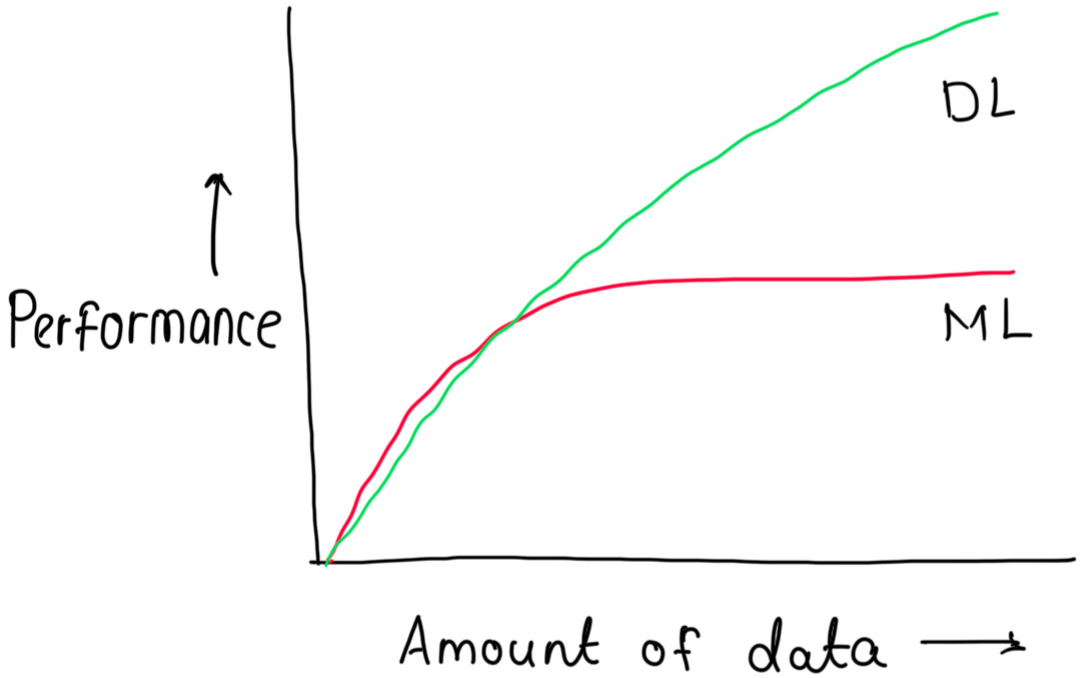
DL algorithms uses multiple layers to progressively extract higher level

Features from the raw input.

For example, in image processing, lower layers may identify edges, while higher layers may identify the concepts relevant to a human such as digits or letters or faces.

Deep Learning Vs Machine Learning

1) Data Dependency



2) Hardware Dependency

DL needs GPU to train models.

3) Training Time

High Training Time

4) Feature selection

Automatically Feature selection

5) Interpretability

Factors Behind Deep Learning

1) Data

2) Hardware

3) Frame works / Libraries

(TensorFlow / Keras / Pytorch)

4) DL Architecture

- Image classification - ResNet
- Text classification - BERT
- Image segment - UNET
- Image translation - Pix 2 Pix
- object detection - YOLO
- Speech Generation - WaveNET

5) People / Community